

jaxon: a differentiable, GPU-native simulator for peripheral-nerve fiber models

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Abstract

jaxon is an open-source, fully differentiable and GPU-native reimplementation of the canonical peripheral-nerve fiber models in JAX/Jaxley: the myelinated McIntyre–Richardson–Grill (MRG) and Sweeney axons and the unmyelinated Sundt and Rattay C-fibers. It reproduces NEURON’s **extracellular** mechanism through a custom backward-Euler coupled intracellular/periaxonal double-cable solver, agreeing with pyfibers-wrapped NEURON on 99.6 % of 943 activation-threshold configurations within 1 % and matching conduction velocity to machine precision. Because the entire forward model is expressed in JAX, it is both vectorized—simulating whole fiber populations in parallel and reaching a geometric-mean $\sim 820\times$ speedup at $N = 100,000$ fibers on a single GPU—and differentiable, so extracellular-stimulation parameters (per-contact amplitudes, waveform shape, and electrode position) can be optimized directly through the cable equation rather than grid-searched. *jaxon* slots into existing peripheral-nerve modeling pipelines as a gradient-enabled, population-scale replacement for the NEURON forward solver.

Keywords: peripheral nerve stimulation, fiber model, differentiable simulation, JAX, GPU, neuromodulation, selectivity optimization

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v1.0.0
C2	Permanent GitHub link to code/repository used for this code version	https://github.com/CellularSyntax/jaxon
C3	Legal Code License	MIT
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	Python 3.11; JAX 0.6.2, Jaxley 0.13.0, Optax 0.2.8, NumPy 2.2.6, SciPy 1.15.3, Matplotlib 3.10.9; NEURON 9.0.1 and PyFibers 0.8.5 (reference/validation only); optional CUDA 12 for GPU execution
C6	Compilation requirements, operating environments & dependencies	Platform-independent (Linux/macOS/Windows-WSL2); a Conda environment (<code>environment.yml</code>) provides all dependencies; an optional NVIDIA GPU (CUDA 12) enables batched GPU execution
C7	If available, link to developer documentation/manual	https://github.com/CellularSyntax/jaxon (README, REPRODUCE.md, AUDIT.md)
C8	Support email for questions	max.haberbusch@meduniwien.ac.at

Table 1: Code metadata.

Required Metadata

Current code version

1. Motivation and significance

Electrical stimulation of peripheral nerves is an established therapy for a growing range of conditions, and selecting where to place an electrode and how to shape the stimulus increasingly relies on computational models. The

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standard modeling approach couples a volume-conductor description of the tissue—typically solved by the finite-element method (FEM)—to biophysical, multi-compartment cable models of individual axons [2, 3]. The extracellular potential computed along each fiber drives the membrane model, and threshold and recruitment follow. The reference implementation of the fiber side of this pipeline is NEURON [7], most conveniently accessed through PyFibers [8], which wraps the canonical published axon models.

This reference stack is accurate but has two structural limitations for modern neural-interface design. First, it is *serial*: NEURON simulates one fiber at a time on the CPU, so scoring an optimized stimulus over an entire fiber population—hundreds to tens of thousands of axons per nerve—is prohibitively slow. To stay tractable, models almost universally reduce each fascicle to one or a few representative fibers, and whether that reduction biases the selectivity they predict has been effectively untestable. Second, it is *non-differentiable*: because there is no gradient of the fiber response with respect to the stimulus, stimulation parameters must be tuned by grid search or gradient-free optimization, both of which scale poorly as the number of contacts and waveform degrees of freedom grows. GPU-accelerated surrogates such as AxonML [11] address the speed problem by learning a fast approximation of the threshold, but a surrogate is a fit to the biophysics rather than the biophysics itself.

jaxon was built to remove both limitations while keeping the biophysics exact. It reimplements the canonical peripheral-nerve fiber models—the myelinated MRG [3] and Sweeney [4] axons and the unmyelinated Sundt [5] and Rattay [6] C-fibers—in JAX [9] on top of the Jaxley differentiable cable simulator [10], and adds a custom backward-Euler solver for the coupled intracellular/periaxonal double-cable equation that is the principled equivalent of NEURON’s `extracellular` mechanism. Because the whole forward model is JAX, it is simultaneously *vectorized*—running whole populations in parallel with `jax.vmap`—and *differentiable*—exposing exact gradients of the fiber response with respect to the stimulus. *jaxon* is validated against pyfibers-wrapped NEURON and reaches a geometric-mean $\sim 820\times$ speedup at $N = 100,000$ fibers on one GPU, fast enough to optimize and then re-score whole-nerve populations. A companion paper [1] uses *jaxon* to show that the common single-representative-fiber reduction overestimates deliverable selectivity in a species-dependent way; the present paper describes the software itself.

2. Software description

2.1. Software architecture

jaxon is organized as a differentiable forward model wrapped by an optimization and analysis layer (Fig. 1). At its center is a vectorized integrator of the coupled (V_i, V_{px}) cable equation that maps a set of per-contact stimulus amplitudes (and, optionally, waveforms and electrode positions) onto the membrane response of an arbitrary number of fibers at once. The model is used through three interchangeable entry points that all act on the same underlying functions: a Python package (**jaxfibers**) for library use, a set of experiment scripts (**experiments_v2**) that reproduce the paper, and SLURM batch drivers for cluster runs.

The package is structured by concern:

- **jaxfibers.channels** — each NMODL ion-channel mechanism hand-translated into a pure-JAX **Channel** class, with rate constants checked against the published **.mod** formulas to machine precision;
- **jaxfibers.fibers** — morphology builders, one per model, returning a **jaxley.Cell** and a geometry dataclass describing the per-compartment static parameters;
- **jaxfibers.stim** — the intracellular and point-source extracellular stimulus helpers, the coupled backward-Euler solver (**extracellular_coupled.py**), the vmapped multi-fiber forward pass (**batch_solve.py**), and the multi-contact ring-cuff field;
- **jaxfibers.optim** — the recruitment/activation proxy, the selectivity index, and the Adam optimization loops.

The coupled solver integrates the double-cable equation by backward Euler; each time step is a 2×2 block-tridiagonal system in the interleaved intracellular and periaxonal unknowns, solved directly by a JAX-compatible block-Thomas sweep (verified to 5.6×10^{-11} mV against a dense LU reference), with back-substitution via **jax.lax.associative_scan** for $O(\log n)$ depth. Because every operation is a JAX primitive, the same forward pass is (i) batched over fibers with **jax.vmap** for population-scale throughput and (ii) differentiated with reverse-mode autodiff (or a packed finite-difference gradient) for stimulus optimization. For the cohort application, per-contact lead fields are supplied by the *golgi* platform [12, 13] from histology-segmented nerve geometries; a lightweight analytic point-source field is included for local demonstrations.

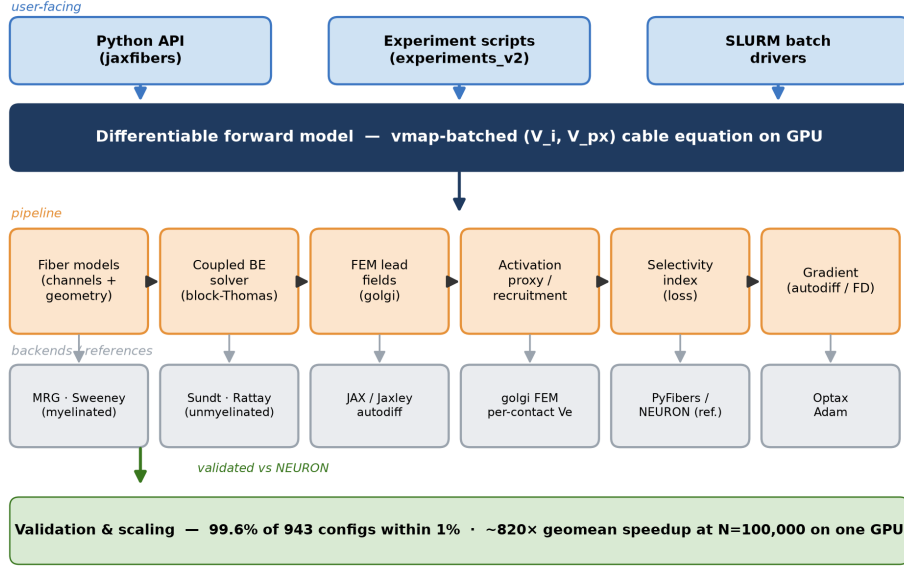


Figure 1: Architecture of *jaxon*. Three user-facing entry points (the *jaxfibers* Python package, the *experiments_v2* scripts, and SLURM batch drivers) share one differentiable, *vmap*-batched forward model of the coupled (V_i, V_{px}) cable equation. The pipeline runs on JAX/Jaxley (autodiff), *golgi* FEM per-contact lead fields, and PyFibers/NEURON as the validation reference; the myelinated (MRG, Sweeney) and unmyelinated (Sundt, Rattay) fiber models, the block-Thomas coupled solver, the activation/selectivity losses, and the gradient form the stages. The whole model is validated against NEURON and scales to $N = 100,000$ fibers on one GPU.

2.2. Software functionalities

jaxon's major functionalities are:

1. **Canonical fiber models.** Four published axon models—myelinated MRG [3] (with both the original and interpolated parameter tables) and Sweeney [4], and unmyelinated Sundt [5] and Rattay [6] C-fibers—reimplemented as pure-JAX channels and morphologies.
2. **Coupled extracellular solver.** A backward-Euler integrator for the two-state (V_i, V_{px}) double-cable equation—the principled equivalent of NEURON's *extracellular* mechanism rather than the single-cable activating-function approximation—solved by a JAX-compatible 2×2 block-Thomas sweep.
3. **Population-scale batched forward pass.** *jax.vmap* packs N fibers into a single vectorized simulation, enabling whole-nerve populations to be simulated and re-scored on one GPU.
4. **Differentiable stimulus optimization.** Three optimization modes: per-contact rectangular amplitudes via a packed finite-difference gradi-

ent ($K+1$ configurations in one forward pass); arbitrary $K \times T$ waveforms via reverse-mode autodiff through the ODE scan (with gradient checkpointing); and joint amplitude-plus-electrode-position search, with the extracellular field recomputed differentially each step.

5. **Selectivity scoring.** A recruitment (activation) proxy and a selectivity index defined as the difference between the recruited fraction of on-target and off-target fibers, usable both as an optimization objective and as a post-hoc population score.
6. **FEM-field and multi-contact support.** Per-contact lead fields from the *golgi* platform over histology-segmented nerves and a multi-contact ring cuff, with a superposition-based current-steering model; an analytic point-source field is provided for demonstrations and tests.
7. **Validation and benchmarking.** Scripts that reproduce the threshold, conduction-velocity, and propagation-phenomena comparisons against pyfibers-wrapped NEURON, and the CPU/GPU scaling benchmark.

2.3. Sample code

The library exposes the forward model and the optimizers directly; a minimal selectivity optimization is a short script:

```
import jax
from jaxfibers.fibers import build_mrg
from jaxfibers.stim import load_lead_fields, batch_forward
from jaxfibers.optim import selectivity_index, optimize_amplitudes

# 1. build a population of MRG fibers for one nerve
pop = build_mrg(diameter_um=5.7, n_fibers=1000)

# 2. per-contact FEM lead fields (golgi) + target fascicle mask
Ve, target_mask = load_lead_fields("duke_Ves/human_sub-50_sam-2")

# 3. optimize the 12 contact amplitudes for on-target selectivity,
#    differentiating through the batched cable solve
amps = optimize_amplitudes(pop, Ve, target_mask,
                           objective=selectivity_index,
                           n_steps=90)

# 4. re-score the optimized stimulus on the full population
acts = batch_forward(pop, Ve, amps)
print("population selectivity:", selectivity_index(acts, target_mask))
```

3. Illustrative examples

Figure 2 summarizes the validation and scaling of *jaxon* against the NEURON reference. Across 943 activation-threshold configurations (multiple diameters, pulse shapes, and pulse widths spanning the four models), 99.6 %

agree with pyfibers-wrapped NEURON within 1 %, and conduction velocity matches to machine precision. The same forward model, batched with `jax.vmap`, scales from single-fiber parity with NEURON to a geometric-mean $\sim 820\times$ speedup at $N = 100,000$ fibers on a single GPU ($\sim 200\text{--}300\times$ at the ~ 1000 -fiber scale typical of one nerve).

Figure 3 shows the capability that motivates the differentiable design: using per-contact FEM lead fields over a 12-contact cuff, *jaxon* optimizes the contact amplitudes for target-fascicle selectivity by differentiating through the batched cable solve, then re-scores the optimized stimulus on the full fiber population. Running the optimization on one representative fiber per fascicle and then deploying to the whole population reveals a species-dependent selectivity penalty; re-scoring on the full population—cheap in *jaxon*—recovers most of it. The companion paper [1] develops this result across a cohort of 29 histology-segmented swine and human vagus nerves.

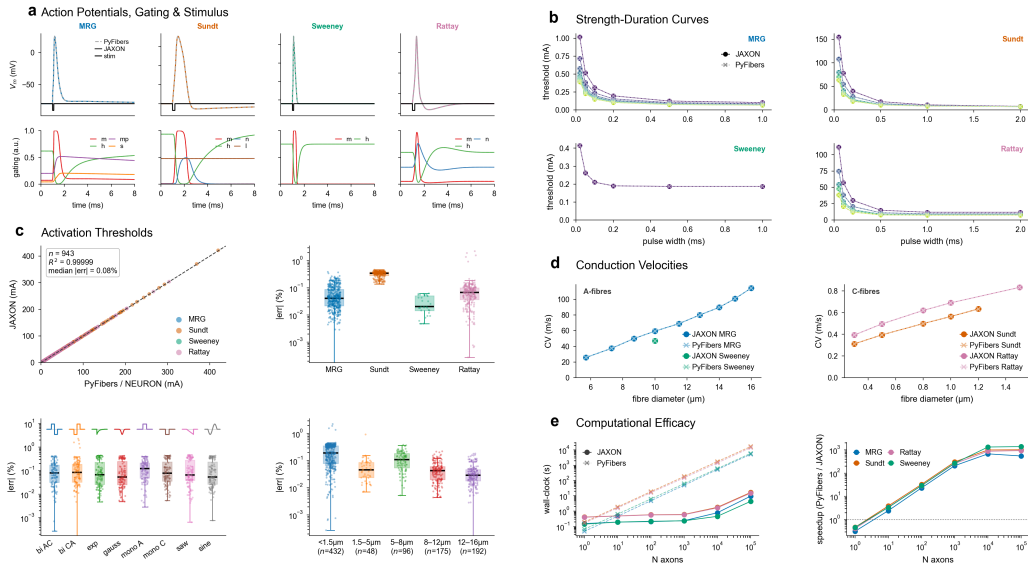


Figure 2: Validation and throughput. *jaxon* (solid) versus pyfibers-wrapped NEURON (dashed) on activation thresholds and conduction velocity across the four fiber models, and the `vmap`-batched CPU/GPU scaling benchmark. 99.6 % of 943 threshold configurations agree within 1 %; the geometric-mean speedup reaches $\sim 820\times$ at $N = 100,000$ fibers on one GPU.

Both examples are reproducible from the released code: the validation and scaling figures regenerate from the archived outputs with a single driver script (`reproduce_figures.sh`), and `REPRODUCE.md` documents the full path from raw simulation to every figure and reported number.

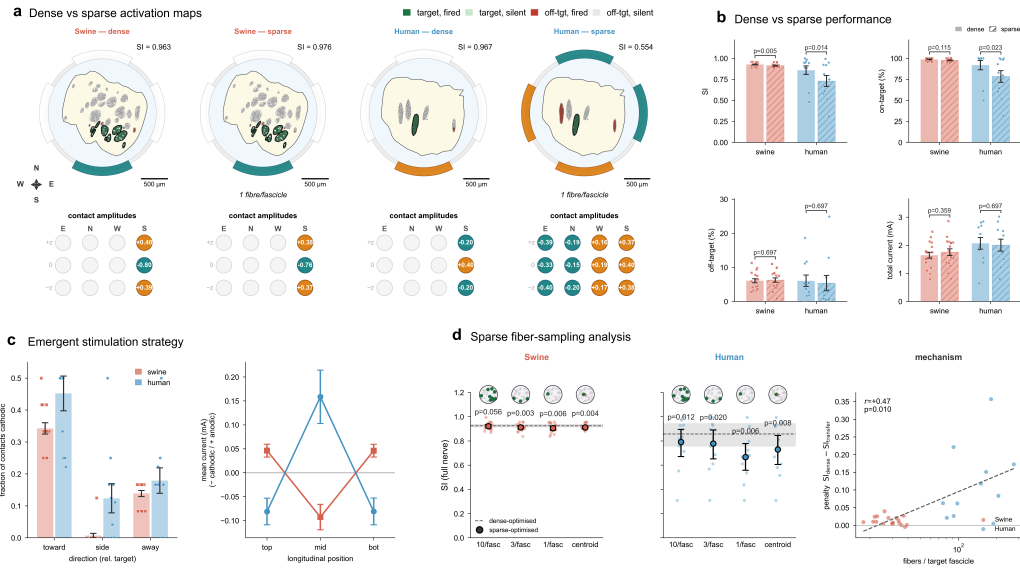


Figure 3: Differentiable selectivity optimization on a histology-segmented vagus nerve. Using *golgi* per-contact lead fields over a 12-contact cuff, *jaxon* optimizes the contact amplitudes for target-fascicle selectivity through the batched cable solve and re-scores the result on the full fiber population. The optimized field, the recruited on- versus off-target fibers, and the resulting deployment penalty are shown for a representative nerve pair; the cohort analysis is developed in the companion paper [1].

4. Impact

jaxon changes what is computationally feasible in peripheral-nerve-stimulation modeling along two axes that the reference NEURON stack leaves closed.

First, it **makes population-scale fiber modeling routine**. By expressing the canonical axon models as a `vmap`-batched JAX forward pass, *jaxon* simulates whole fiber populations on a single GPU at a geometric-mean $\sim 820\times$ speedup over serial NEURON, turning a previously prohibitive whole-nerve re-scoring into a step cheap enough to run inside an optimization loop. This directly enables the central question of the companion study [1]—whether reducing each fascicle to a representative fiber biases predicted selectivity—which was untestable precisely because the population simulation was too expensive.

Second, it **makes the stimulus itself an optimization variable**. Because the whole model is differentiable, per-contact amplitudes, arbitrary waveforms, and electrode positions can be optimized by gradient descent through the exact cable equation, rather than by grid search or a learned surrogate. Unlike a surrogate model, the gradient is of the biophysics, not of a fit to it, so an optimized stimulus can be validated in the same tool that produced it.

Third, it **drops into existing pipelines with minimal friction**. *jaxon* is validated against the community-standard PyFibers/NEURON reference (99.6% of 943 configurations within 1%; conduction velocity to machine precision) and released open-source under a permissive license, so it can serve as a gradient-enabled, population-scale replacement for the NEURON forward solver wherever fiber-level thresholds are needed. Its per-contact-lead-field interface connects it to FEM tooling such as *golgi*, positioning it as an interoperable component rather than a standalone silo.

By combining exact, validated biophysics with GPU-scale throughput and end-to-end differentiability, *jaxon* lets neural-interface researchers ask and answer optimization and population-scale questions that the established serial, non-differentiable tools cannot.

5. Conclusions

jaxon is an open-source, differentiable, GPU-native reimplement of the canonical peripheral-nerve fiber models—MRG and Sweeney (myelinated) and Sundt and Rattay (unmyelinated)—built on JAX/Jaxley with a custom coupled (V_i, V_{px}) backward-Euler solver. It matches pyfibers-wrapped NEURON on 99.6% of 943 threshold configurations and to machine precision in conduction velocity, scales to a geometric-mean $\sim 820\times$ speedup at

$N = 100,000$ fibers on one GPU, and exposes exact gradients of the fiber response for stimulus optimization. Validated against NEURON and applied to a species-dependent selectivity question in the companion paper [1], *jaxon* provides a gradient-enabled, population-scale replacement for the NEURON forward solver in peripheral-nerve modeling pipelines.

Data availability and licensing

jaxon is released as open-source software under the MIT license (code metadata C3/S3). The source code is available on GitHub (<https://github.com/CellularSyntax/jaxon>), and the tagged v1.0.0 release is archived on Zenodo [14]. The processed data required to reproduce every figure and reported number—the *golgi*-derived per-contact lead fields and the validation, scaling, and cohort-sweep outputs—is archived as a companion Zenodo data record [15] under the Creative Commons Attribution 4.0 International (CC-BY-4.0) license, distinct from the software license. The raw finite-element meshes are *golgi* outputs and are available through the *golgi* platform [12, 13] and the underlying SPARC datasets. Reproduction is documented in `REPRODUCE.md` in the repository.

CRedit authorship contribution statement

David Lung: Software, Investigation, Validation, Writing – review & editing. **Max Habermusch:** Conceptualization, Methodology, Software, Supervision, Funding acquisition, Writing – original draft, Writing – review & editing.

Acknowledgements

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Current executable software version

Nr.	(Executable) software meta-data description	Please fill in this column
S1	Current software version	v1.0.0
S2	Permanent link to executables of this version	Zenodo, https://doi.org/10.5281/zenodo.XXXXXXX (v1.0.0)
S3	Legal Software License	MIT (matches C3)
S4	Computing platforms/Operating Systems	Linux, macOS, Windows (WSL2)
S5	Installation requirements & dependencies	Conda environment (<code>environment.yml</code>); JAX 0.6.2, Jaxley 0.13.0, Optax; NEURON 9.0.1 + PyFibers 0.8.5 for the validation reference; optional NVIDIA GPU + CUDA 12 for batched GPU execution
S6	If available, link to user manual	https://github.com/CellularSyntax/jaxon
S7	Support email for questions	max.haberbush@meduniwien.ac.at

Table 2: Software metadata.